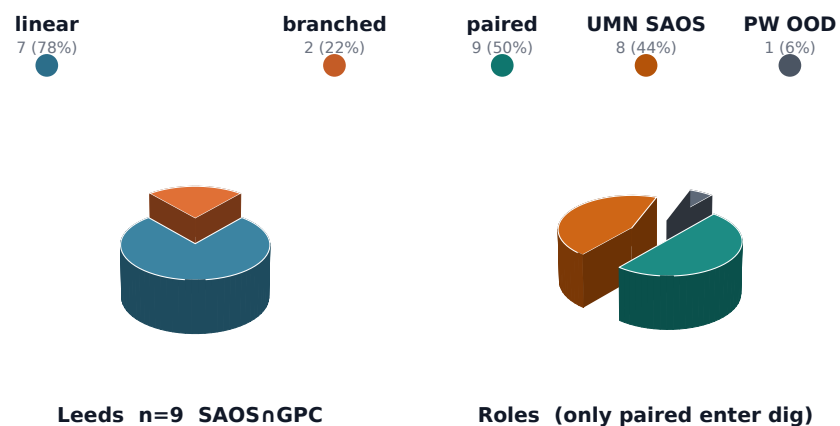
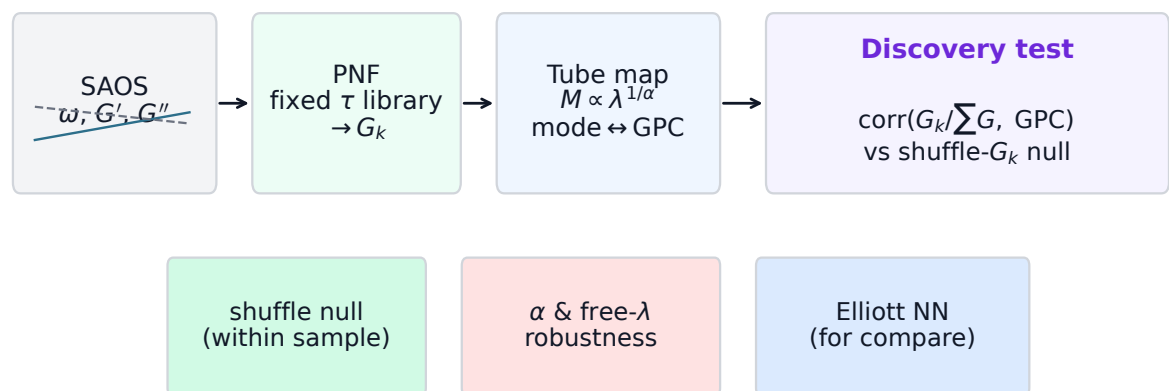


Readable PNF amplitudes are molecularly aligned with MWD

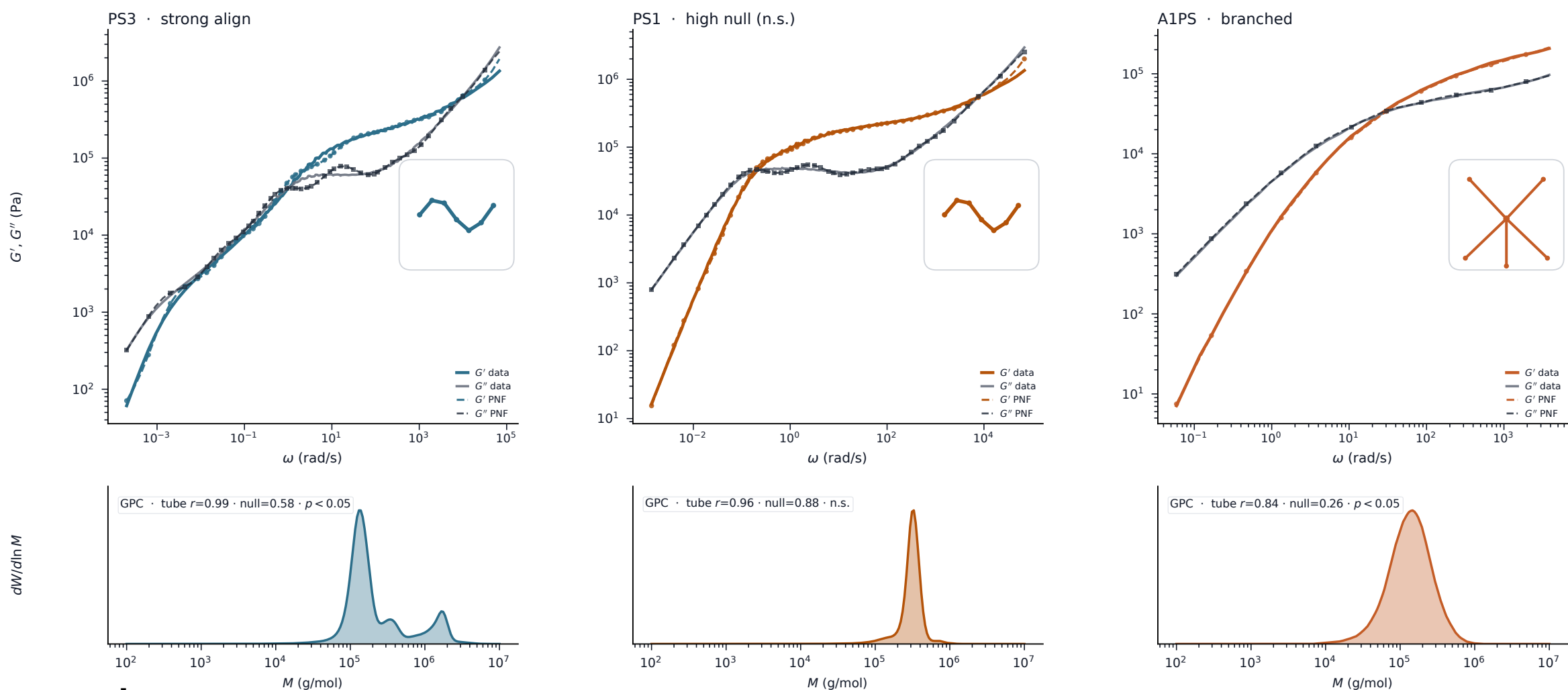
a Cohorts — paired melts vs external probes



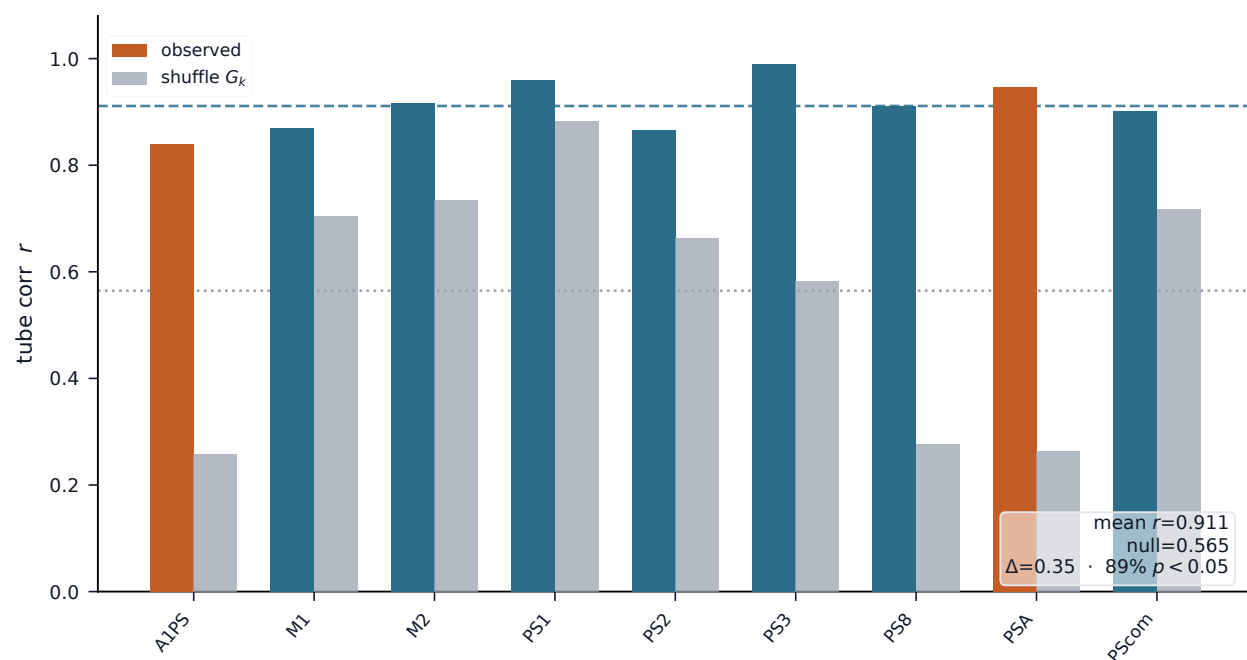
b From SAOS master curves to tube-aligned PNF mode mass



c Key cases — SAOS + PNF reconstruction + GPC (topology inset)



d Outcome — tube alignment $\gg$ shuffle null



e Robustness — α , free- λ , bin proxy

